520.681: Perception for Embodied AI
Homework 1 - Written
Due: Feb 18, 2026, 11:59pm ET

Instructions

- This is individual homework and worth 100 points.
- You must submit your solutions on Gradescope (you should already be automatically enrolled via Canvas).
- We recommend that you use LaTeX, but we will accept scanned solutions as well.

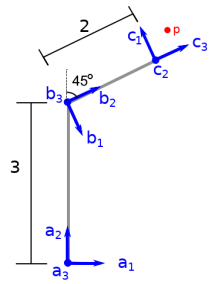


Figure 1: Robot arm diagram. Note that a_3, b_3, and c_2 are pointed out of the page.

1 Homogeneous Transforms (20 points)

Consider the robot arm in Figure 1. There are three coordinate frames A, B, C with their corresponding basis vectors. Let ${}^{A}p$ denote the coordinates of the point p in frame A, ${}^{B}p$ be the same point in frame B, and ${}^{C}p$ the same point in frame C. Define the homogeneous transforms T_1, T_2 , T_3 as follows:

$$\begin{bmatrix} {}^{A}p \\ 1 \end{bmatrix} = T_1 \begin{bmatrix} {}^{B}p \\ 1 \end{bmatrix} \qquad \begin{bmatrix} {}^{B}p \\ 1 \end{bmatrix} = T_2 \begin{bmatrix} {}^{C}p \\ 1 \end{bmatrix} \qquad \begin{bmatrix} {}^{A}p \\ 1 \end{bmatrix} = T_3 \begin{bmatrix} {}^{C}p \\ 1 \end{bmatrix}$$

1. Find the homogeneous transform T_1. (6 points)
2. Find the homogeneous transform T_2. (6 points)
3. Find the homogeneous transform T_3. (Hint: Use T_1 and T_2). (8 points)

2 Rotation Matrix Sudoku (10 points)

Only three elements of the rotation matrix below are known (entries with $*$ are unknown). Use the properties of rotation matrices to determine the missing elements. Either prove your solution is unique, or provide an alternative solution. (*Note: Show your work to receive partial credit. You may show your math or include a code segment.) [1]

$$\begin{bmatrix} * & 0.892 & 0.423 \\ * & * & * \\ -0.186 & * & * \end{bmatrix}$$

[1] Given the computational nature of this problem, you should use this opportunity to get comfortable doing calculations in Python 3 using PyCharm's interactive console. You'll find Numpy arrays invaluable, including the associated Numpy functions for 'inv', 'cross,' 'dot,' and matrix multiplication using the '@' symbol.

3 Rodrigues' Formula (10 points)

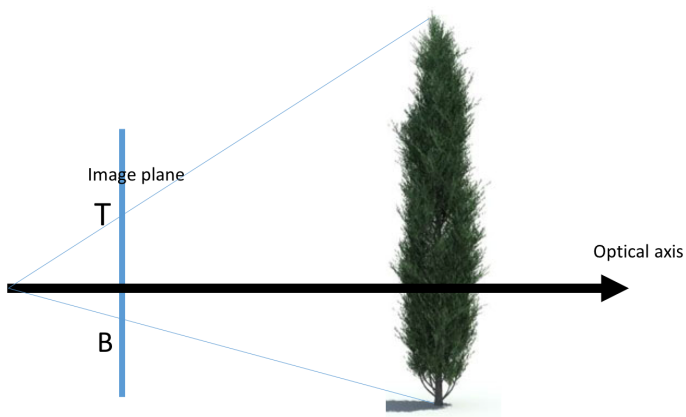
Write the axis-angle representation and the quaternion corresponding to the rotation matrix. (10 points)

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & -\cos(\pi/6) & \sin(\pi/6) \\ 0 & \sin(\pi/6) & \cos(\pi/6) \end{pmatrix}$$

4 Perspective Projection (no need for homogeneous coordinates) (20pts)

1. (10pts) A world point with camera coordinates (300, 600, 1200) is perspectively projected into an image at coordinates (55, 110). Given that the image center is at (0,0) and the aspect ratio (ratio between width and height) of the pixels is 1, what is the focal length f of the camera?

2. (10pts) Assume that you see the bottom and the top of a vertical tree in front of you. The image plane is vertical as well with $f = 1$ and you see the bottom and the top of the tree at calibrated coordinates $B = (0, y_1)$ and $T = (0, y_2)$, respectively. w Without knowing the tree's height can you tell whether the tree



will hit the projection center if it falls? Prove your answer.

5 Projective geometry & Homographies (30 pts)

1. (4 pts) Consider homogeneous coordinates $[X, Y, Z]$, what is the point represented when $Z \neq 0$? What about $Z = 0$?

2. (4pts) For each of the following pairs of points in $\mathbb{P}^2$, write down an equation for the line that passes through them:

$$(a)\ [-2, 5, 3] \text{ and } [1, 3, 4]$$
$$(b)\ [a, 0, b] \text{ and } [0, c, b]$$
$$(c)\ [a, 0, 0] \text{ and } [0, 0, a]$$

3. (4 pts) For each of the following pairs of lines in $\mathbb{P}^2$, determine the point of intersection:

$$(a)\ 3x - y + 2w = 0 \text{ and } x + 5y - w = 0$$
$$(b)\ 2x - 6w = 0 \text{ and } 5x - 2y = 0$$
$$(c)\ 7x + y - w = 0 \text{ and } w = 0$$

4. (4 pts) Given the two lines $l_1 := w = 0$ and $l_2 := x + 2y + w = 0$ in $\mathbb{P}^2$, for each of the following definitions of l_3, determine if there is a common intersection between l_1, l_2 and l_3 and if so, what that common intersection point is.

$$(a)\ l_3 := x + 2y + 6w = 0$$
$$(b)\ l_3 := -3x - 6y + 6w = 0$$
$$(c)\ l_3 := 7x + y - w = 0$$

5. (4 pts) Find a projective transformation T that maps the points $[1, 0, 0]$, $[0, 1, 0]$ and $[0, 0, 1]$ to the non-collinear points $[1, -1, 1]$, $[1, -2, 2]$ and $[-1, 2, -1]$, respectively.

6. (4 pts) A homography H leaves horizontal lines horizontal but maps all vertical lines to lines going through the point $(0, 3)$. It leaves the points $(0, 0)$ and $(1, 1)$ at the same place. Compute H.

7. (6pts) Find a condition on the elements of a homography H such that parallel lines remain parallel. Such a transformation is called an affine transformation.

6 Vanishing Points & Horizon (10pts)

The image of the rectangular facade of a building has two vanishing points, one at $(-b, 0)$ corresponding to horizontal lines and one at $(0, h)$ corresponding to vertical lines. Assume that the origin $(0, 0)$ and the point $(1, 1)$ remain fixed.

1. (5 pts) Find the transformation that will map the facade to a rectangle.
2. (5 pts) Compute the horizon (the image of the line at infinity).